Data 601 Final Project.

Predicting and Analyzing Shipping Within E-Commerce.

Dawson Conlon

Ricky Doucette.

Introduction.

E-Commerce is the buying and selling of goods online, within the domain, one of the most important factors of having a successful business is shipping. It is valuable to understand how efficient shipping is within these companies and what predictions can be made around shipping when utilizing machine learning models, such as predicting how much shipping is going to cost. The questions to be analyzed is whether we can predict shipping costs and whether the number of delivery days impacts shipping costs. The desired outcome is to create a robust model that can predict shipping costs.

The data set was an open source synthetically generated E-Commerce data set from Kaggle. The data had over 150,000 rows of customer transactions and the largest table had 46 columns. The data types of the features we used in our model are listed below.

Category	Column
Numerical	Estimated Delivery Days
Numerical	Delivery Days
Numerical	Quantity
Numerical	Profit
Numerical	Gross Sales
Numerical	Discount Amount
Numerical	Tax Amount
Numerical	Shipping Cost
Numerical	Net Sales
Numerical	Product Cost
Numerical	Customers Lifetime Value
Numerical	Customer Rating
Numerical	Customer Order Count
Identifier	Order ID

The website where this data was collected doesn't provide information on its synthetic generation process. Since it was synthetically created, its accuracy and validity are difficult to assess. However, given its size, this dataset was likely generated by a large language model (LLM). The inherent patterns in synthetic data are designed by the author and the prompts used on the LLM, which diminishes its real world validity. We analyzed a random sample of 100 customer interactions from the store's 150,000 interactions. This data is suitable for a supervised learning regression problem, as all relevant features we want to predict are numerical. Therefore, we employed a linear regression model to predict shipping costs.

Exploratory Data Analysis

We wanted to get an understanding of the distribution of shipping cost, this would allow us to see how much customers are paying for shipping usually.



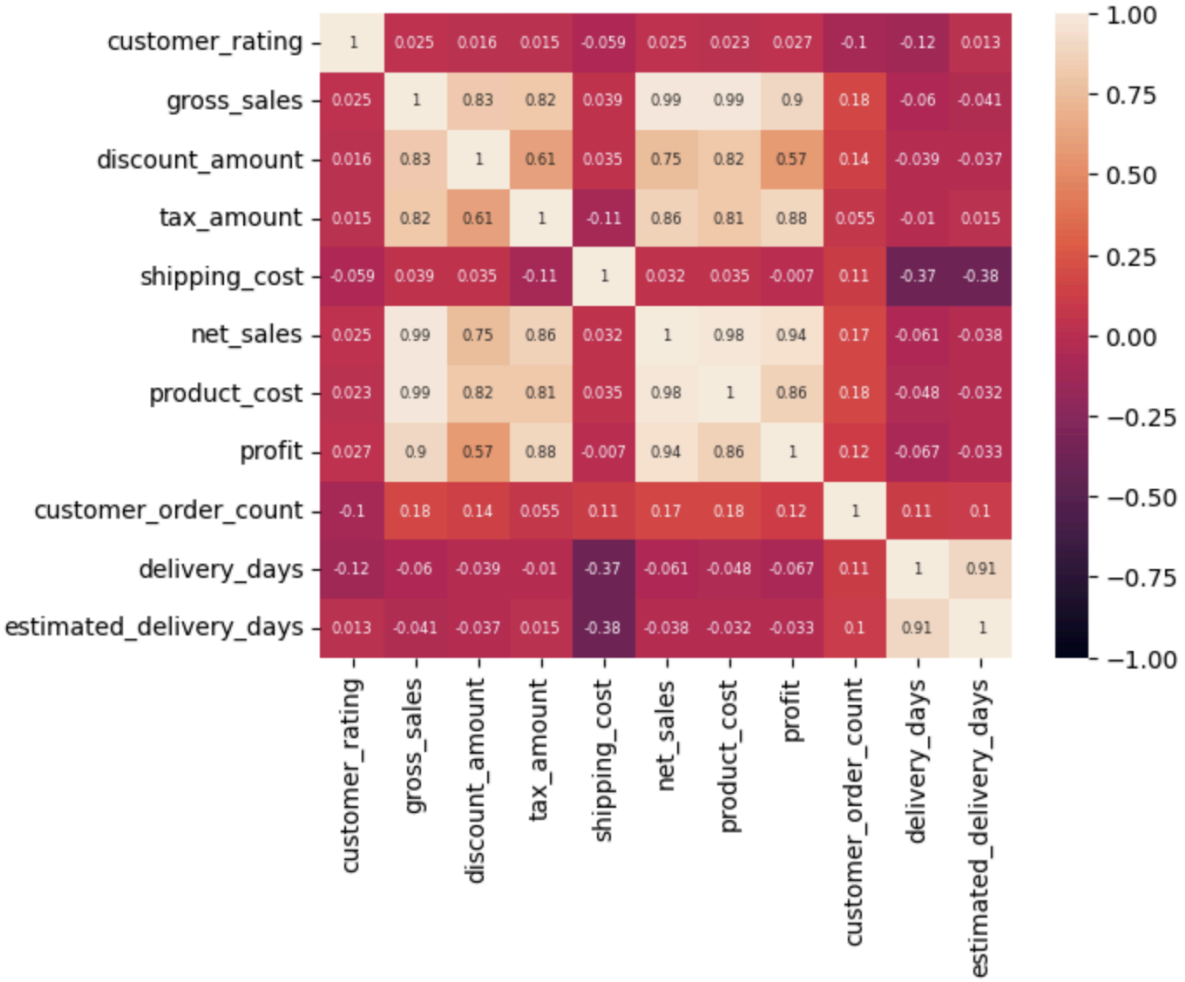
Based on the graph above we can see that there is a right skew to the distribution of shipping cost and customers are usually paying $0 to $8 for shipping. This was further supported when the mean (7.8$) and the median ($5.8) was calculated for shipping cost based on the sample of 100 customer interactions.

We then wanted to see how the amount of time it takes to get an order delivered to you impacts the shipping cost. We took a look at the variables delivery days and shipping cost and made a scatter plot to understand the relationship of these two variables.



We can see in the scatterplot that there's many different shipping costs for the same value of delivery days and there doesn't appear to be any kind of pattern. What this shows us is that there seems to be no relationship between shipping cost and delivery days. Customers will tend to pay different shipping costs even if they receive their orders in the same amount of days as another customer.

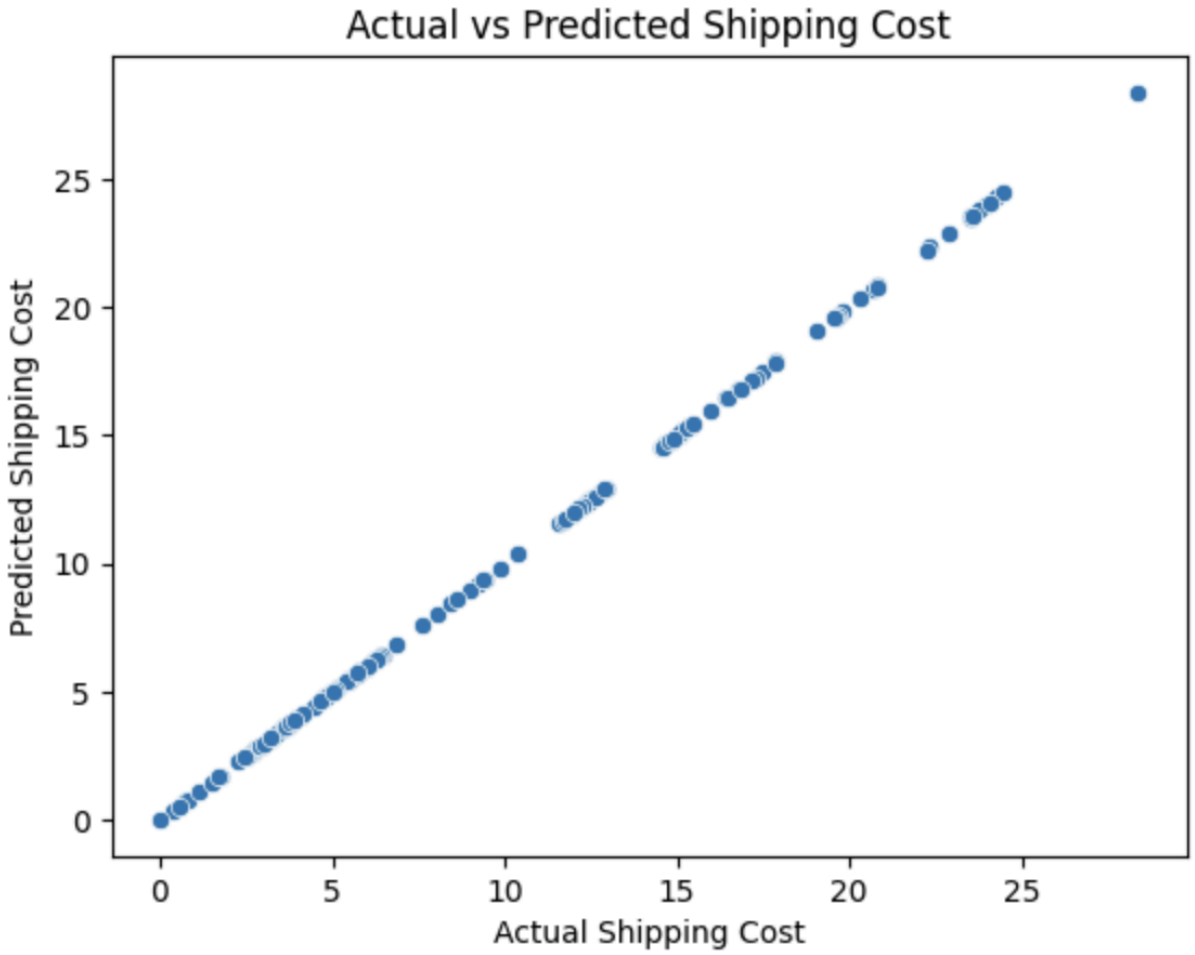
We wanted to look at a correlation heatmap of the features we planned to use in the model for shipping cost to help determine if there's a linear relationship with shipping cost and any other features of interest.



We can see shipping cost doesn't seem significantly correlated with anything. There is a small negative correlation between shipping costs and delivery days / estimated delivery days but that is it. So when shipping cost goes up, delivery days tend to go down. We can also see that most of the other features that are used to calculate shipping cost are highly positively correlated with each other, besides delivery days, estimated delivery days, and customer rating. There's also the obvious highly positive correlation between delivery days and estimated delivery days.

The Model

To prepare the linear regression model we first did an 80/20 split between the training data and the testing data. We made sure our variable to be calculated, shipping cost, was not amongst the features used to build the model and instead was being used as the feature to be predicted. We got our R squared score of exactly 1.0. To see if it changes we tried different splits of training and testing data, but the score did not change. What this tells us is that 100% of the variation in shipping cost can be predicted by our mode. This could be due to overfitting, but the R squared score of the test data as well as the training data were both 1.0 exactly so we suspected there to be a mathematical formula to perfectly calculate shipping cost using some of the features used to create the model. We also got our mean absolute error value which was so small it was basically 0. This is supporting that our model perfectly predicts shipping cost.



As we can see this graph supports our findings of our model having an R squared of exactly 1.0. There is a perfect straight line of all the predicted shipping cost vs the actual shipping cost.

We then got the correlation of all the features with shipping cost and none had a perfect score of 1.0, so one single feature isn't responsible for the entirety of the prediction of shipping cost.

	Shipping_cost correlation
estimated_delivery_days	-0.290932
delivery_days	-0.257004
tax_amount	-0.064597
customer_rating	-0.033755
discount_amount	-0.028267
customer_order_count	-0.022938
product_cost	-0.015002
gross_sales	-0.011062
customer_lifetime_value	-0.003426
profit	-0.002700
net_sales	0.001026
quantity	0.023580

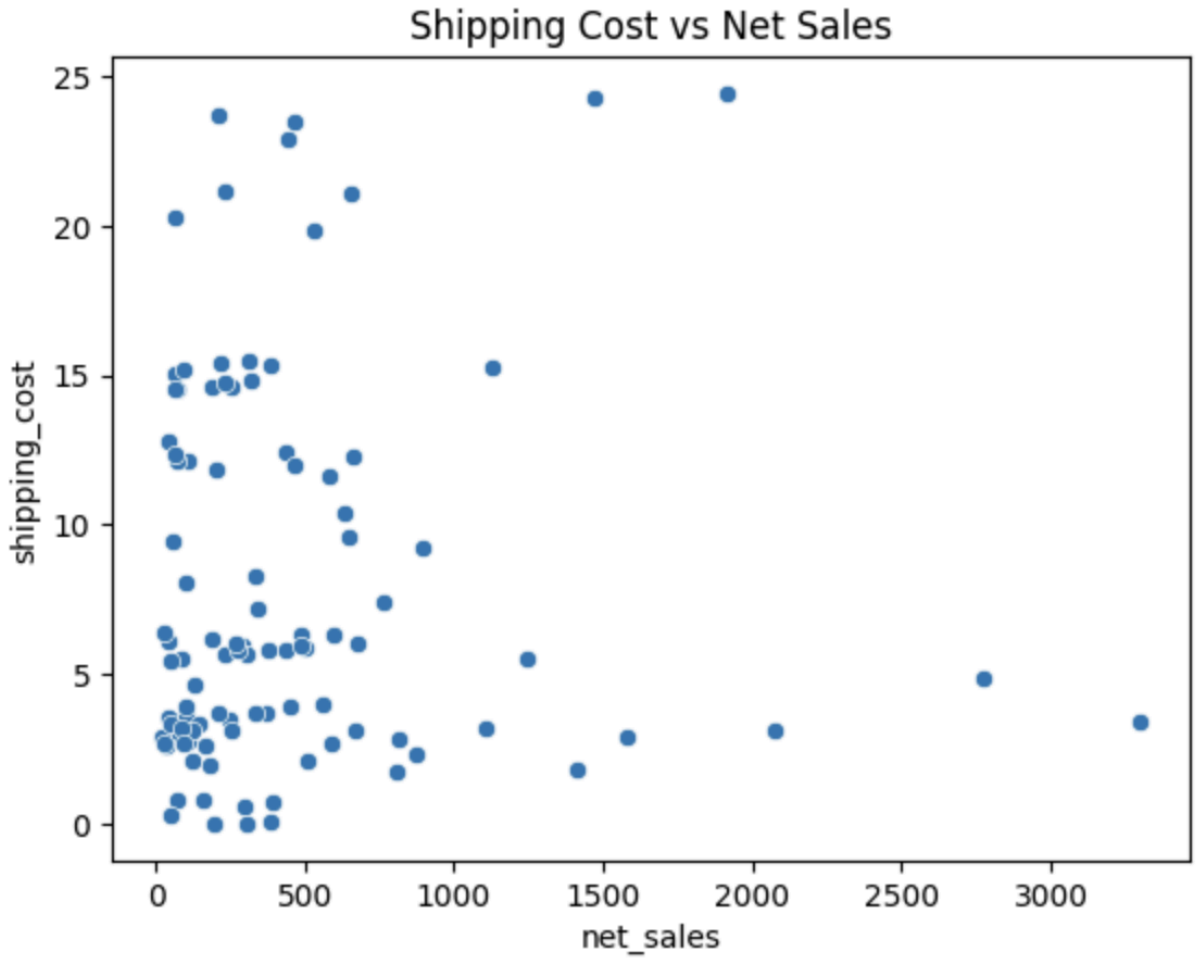
Following that we got the coefficients of the features used in our linear regression and could determine from their coefficients being negligible that half the features weren't used in the calculation of shipping cost. The other half were clean values with only one decimal point. The values were either 0.4, -0.4, or 0.6. Net sales had a coefficient of exactly 1.0.

Features	Coefficient
net_sales	1.0
discount_amount	0.4
customer_rating	4.432950e-14
quantity	3.959906e-14
customer_order_count	6.904019e-15
customer_lifetime_value	1.062518e-17
delivery_days	-8.410190e-17
estimated_delivery_days	-1.747581e-15
gross_sales	-0.4
tax_amount	-0.4
profit	-0.6
product_cost	-0.6

Then to determine which features are the most important in the prediction of shipping cost we got a permutation of importance table. Here we can see that the features that had negligible coefficients in the linear regression have an importance value of zero. This supports our findings that 6 of the features incorporated into the model are used to mathematically calculate shipping cost while the other 6 features aren't used at all. We see that net sales is very important in the model used to predict shipping cost.

Features	Importance
net_sales	15202.742766
gross_sales	3570.215195
product_cost	2686.481317
profit	783.575767
discount_amount	389.201040
tax_amount	25.526169
customer_lifetime_value	0
customer_rating	0
delivery_days	0
estimated_delivery_days	0
customer_order_count	0
quantity	0

Because we saw such a high value of importance for net sales compared to the other features, we graphed shipping cost against net sales to see if there was any linear relationship that we could visibly see.



Based on the scatter plot between just net sales and shipping cost, without having created the model that we did, one would infer that there isn't a relationship between the two. We can see that the vast majority of the data points are at the lower end of net sales and most data points are at the lower end of shipping cost. However, again we don't see any sort of relationship. We do know however that net sales was very important in our model's prediction of shipping cost.

We tested the performance of a model to predict shipping cost using all the same features except net sales. It performed extremely poorly with an R squared of 0.06505853359321212. From here we could have and should have tested other models for prediction of shipping cost excluding one feature at a time that was used in our original model, however we did not do that.

Conclusion

We did succeed in creating a model to calculate shipping cost, however we did not achieve our desired outcome. Our created model mathematically uses a handful of features to directly calculate our feature of interest, shipping cost, and delivery days was not used in this calculation to get shipping cost when we suspected it would be important. This wasn't our desired outcome. We wanted to create a model that accurately predicts shipping cost but not one that perfectly calculates it. Our model however, even with its R squared value of 1.0 on test data, is not overfitting. If given new data, it will perfectly predict shipping cost due to the mathematical relationship. Our next step would be to create a new model with some of the features that are used to directly calculate shipping cost dropped. We started doing this by dropping net sales but our models performance dropped significantly so more altering of the model is required. What we essentially discovered was a model is not required for shipping cost as it can be simply mathematically calculated. A model is not always needed.